Software Requirements Specification for Squigglr: subtitle describing software

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Revision History

Date	Version	Notes
Date 1	1.0	Notes
Date 2	1.1	Notes

1 Purpose of the Project

This section states the problem Squigglr addresses, who has it, and the goals the project commits to. The problem and goals come from the Problem Statement and Goals (PSG) document (SRC-001).

1.1 User Business

Visual-motor integration is the ability to coordinate what a person sees with how they move. Children rely on it when drawing, copying shapes and learning to write (SRC-001). Kindergarten and primary school learners need chances to practise these skills at a suitable level and to complete activities with less help over time. Older children who need continued support, including some children with autism, attention-deficit/hyperactivity disorder (ADHD) or other developmental or learning differences, share this need. A diagnosis alone does not determine the support a child requires (SRC-001).

The adults around these children have a related problem. Parents, guardians, caregivers and educators need to understand which difficulties a child met and how much guidance the child needed. A finished drawing alone does not show whether the child completed it independently, relied on tracing guides, or corrected mistakes during the attempt (SRC-001). Squigglr is a website, used on a touchscreen device with a stylus, that offers guided drawing practice, creative drawing, and records that help adults understand that practice. It makes no diagnosis or treatment claims (SRC-001).

The gap left by current alternatives, as the PSG frames it, is that a drawing's final result is kept without the process behind it: the guidance used, the difficulties met and the corrections made. Squigglr's value is in keeping that process visible to supporting adults while children practise with gradually less support. The comparison with existing products appears in the Off-the-Shelf Solutions section of this document.

1.2 Goals of the Project

Table 1 lists the six project goals, G.1 to G.6, using the PSG's goal titles (SRC-001). Each goal is stated as Purpose, Advantage and Measurement: what the goal is, the benefit it brings and to whom, and how the team will check that it is met. Following the teaching assistant's (TA's) request for

measurable goals (SRC-003) and the supervisor's direction that no requirements are gathered from children (SRC-004), every check is done by the team or by adult reviewers using prepared or adult-made attempts, never with child participants. Where a check needs a number that has not yet been agreed, it uses a named constant marked TBD. Each lesson (for example, drawing a square) moves through fixed stages that give gradually less help: a demonstration, tracing a full outline, drawing between endpoints only, and drawing with no guide. This sequence follows the activity the team presented to the domain specialists (SRC-005, Moustafa, 6:03).

G.3 was revised after the PSG was submitted. The submitted PSG said Squigglr adjusts guidance and difficulty automatically. Following the domain-specialist interview (SRC-005 (P1, 28:37); SRC-005 (P2, 29:18)), and as agreed at the TA meeting of Oct 7 (SRC-006), the team changed it to adult-controlled progression and updated the PSG to match (SRC-001).

Table 1: Project goals

Goal	**Purpose, advantage and measurement**
G.1 Support structured drawing practice	*Purpose:* every lesson in the agreed catalogue covers lines, shapes or simple drawings, starts with a demonstration, and gives feedback and a retry at each drawing stage (SRC-001). *Advantage:* structured practice at a suitable level supports visual-motor development (SRC-001), and children can enter a lesson, practise, restart and go back and forth on their own (SRC-005 (P1, 34:13)). Serves children. *Measurement:* met when the team completes every stage of every catalogue lesson with prepared correct attempts, each attempt passes, and each drawing stage accepts a drawing, gives feedback and allows a retry.

Goal	Purpose, advantage and measurement
G.2 Support progression towards independent drawing	*Purpose:* within a lesson, the child moves to the next stage when an attempt meets that stage's pass criteria (`STAGE_PASS`, TBD); an attempt that does not meet them is not passed, and the child can retry. This checks the drawing against fixed criteria and does not assess the child (SRC-001). *Advantage:* children need to perform activities with gradually less help (SRC-001); a child practises one level until proficient before moving on (SRC-005 (P1, 28:37)), and children reach these skills at different points (SRC-005 (P1, 20:27)). Serves children and supporting adults. *Measurement:* met when (a) every prepared correct attempt advances to the next stage; (b) every prepared incorrect attempt, such as one with a missing side or a wrong endpoint, is not passed and its predefined difficulty is reported; and (c) every saved attempt records the stage it was made at, with no mislabelled record.
G.3 Let supporting adults control support and difficulty	*Purpose:* adults choose which lesson a child practises and may let the child start at a later stage. Squigglr may suggest returning to an earlier stage after `DIFFICULTY_STREAK` (TBD) failed attempts in a row; the change applies only when an adult accepts it, and a declined or ignored suggestion changes nothing. To avoid endless repetition, the lesson ends after `ATTEMPT_LIMIT` (TBD, larger than `DIFFICULTY_STREAK`) unsuccessful attempts at one stage, and the furthest stage reached is saved. *Advantage:* judging readiness automatically would need assessment that is out of scope; the specialists advised adult control (SRC-005 (P1, 28:37); SRC-005 (P2, 29:18); SRC-006). Serves children and supporting adults. *Measurement:* met when, for prepared sequences of successful, unsuccessful and mixed attempts, every suggestion shown matches the agreed criteria, each lesson ends exactly at `ATTEMPT_LIMIT` with its progress saved, and no lesson or starting stage changes without an adult's action.

Goal	Purpose, advantage and measurement
G.4 Support editable creative drawing and refinement	*Purpose:* in the creative space, Squigglr refines a child's drawing of a shape or object in the agreed supported set by adjusting the child's own strokes; the result stays editable and the original can be restored (SRC-001; SRC-005, Moustafa, 6:03). *Advantage:* creative drawing gives room for self-expression, and editability keeps the child in control of the work (SRC-001). Serves children. *Measurement:* using prepared drawings of each item in the supported set, met when for every drawing (a) every point of the refined drawing lies within `REFINE_TOLERANCE` (TBD) of the original strokes; (b) the refined parts can be moved, resized, recoloured, erased and undone; (c) restoring returns the original strokes exactly; and (d) a failed refinement request leaves the drawing unchanged.
G.5 Preserve meaningful activity records	*Purpose:* each attempt's record keeps the stage, every predefined difficulty observed, later corrections and the final outcome; each lesson's record keeps the furthest stage reached and the successful attempts at each stage (SRC-001). *Advantage:* a finished drawing alone hides how much guidance was used and the corrections made (SRC-001). Serves supporting adults. *Measurement:* met when, for prepared attempts and lessons with known actions, stages and outcomes, every saved record matches its known contents, with no difficulty lost after a correction.

Goal	Purpose, advantage and measurement
G.6 Make progress understandable to supporting adults	*Purpose:* summaries show a child's progress over time, separate attempts made without a guide from attempts made with one, and link to the attempts behind them (SRC-001). *Advantage:* adults need to see changes in independent drawing to decide on further practice (SRC-001); the specialists suggested giving the adult a view of the child's progress alongside control of activities (SRC-005 (P2, 29:18)). Serves parents, guardians, caregivers and educators. *Measurement:* met when, for prepared histories showing improvement, recurring difficulties and changing stages, every summary separates the two kinds of attempt correctly and links to the right attempts; and at least `ADULT_REVIEWERS` (TBD) adult reviewers interpret example summaries and locate the supporting attempts, with at least `REVIEW_PASS_RATE` (TBD) of their tasks succeeding.

Table 2 lists the constants used in the goals. Their values are not yet agreed; they will be set with the supervisor and domain specialists and recorded here, and no value will be chosen from child testing (SRC-004).

Table 2: Constants used in the goals

Constant	Meaning	Value
`STAGE_PASS`	Criteria an attempt must meet to pass each stage of a lesson (G.2)	TBD
`DIFFICULTY_STREAK`	Failed attempts in a row before Squigglr may suggest returning to an earlier stage (G.3)	TBD
`ATTEMPT_LIMIT`	Unsuccessful attempts at one stage before the lesson ends and its progress is saved; larger than `DIFFICULTY_STREAK` (G.3)	TBD
`REFINE_TOLERANCE`	Maximum distance between a refined drawing and the original strokes (G.4)	TBD
`ADULT_REVIEWERS`	Minimum number of adult reviewers in the G.6 check	TBD
`REVIEW_PASS_RATE`	Minimum share of G.6 reviewer tasks that must succeed	TBD

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Insert your content here.

2.2 Customer

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26 Ideas for Solution

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27 Appendix — Requirement Sources

This appendix is the requirement sources register. Each source has a fixed identifier (SRC-001 to SRC-007) that is never renumbered. A requirement names the sources it comes from in its rationale or source field. A statement from the specialist interview is cited by its source identifier, participant code and transcript timestamp, for example "(SRC-005, P1, 28:37)"; the statement identifiers in Section 27.1 (for example SRC-005.01) can be cited instead. The interview is expert-reported evidence only. The three participants are de-identified domain specialists (P1, P2 and P3). No child took part in the interview or in any other source listed here.

Table 3 lists the project documents and meeting sources, and Table 4 lists the interview and peer-review sources. The supported requirement identifiers are not listed in these tables yet. They are added as back-links as each requirement is written, so that every dependency appears at both ends: the requirement names its source, and the source lists the requirement. Each source and statement has a LaTeX label (for example `src:005.01`), so a requirement can link to it and the link works in the PDF.

The sources represent users in different ways. Children are the main beneficiaries, but by the supervisor's direction (SRC-004) no requirement is gathered from children and no information about them is collected. Their needs enter the requirements only through adults who know them. The domain specialists (SRC-005) are the only outside voice so far. Their advice changed the project: progression is chosen by an adult instead of adjusted automatically, the stylus is the main input with finger input as a stretch goal, and camera tracing became a stretch goal. Parents, educators and occupational therapists have not been consulted yet. The specialists advised doing so (SRC-005.03, SRC-005.06), and until then the requirements that depend on their views rest on the expert statements below and are open to change.

Table 3: Requirement sources: project documents and meetings

ID	Type	Date	Reference	Summary	Limitations
SRC-001	Project document	–	`docs/Problem StatementAnd Goals/ Problem Statement.tex`	Problem, stakeholders, goals and stretch goals.	Written by the team; some assumptions changed after SRC-005.
SRC-002	Project document	–	`docs/ Development Plan/ Development Plan.tex`	Team process, tools and schedule.	Written by the team; plans, not user needs.
SRC-003	TA meeting	Sep 23, 2026	Issue #10	Website instead of an installed app; teachers and students as primary stakeholders; specific, measurable metrics.	Short minutes; not a user source.
SRC-004	Supervisor meeting	Sep 23, 2026	Issue #21	Collect no information about children; do not gather requirements by asking children; ethics approval for studies; start with a web app; traceability.	Advice from the client, not observed user data.

Table 4: Requirement sources: interview and reviews

ID	Type	Date	Reference	Summary	Limitations
SRC-005	Specialist interview	Oct 1, 2026	Appendix 28	Three de-identified specialists on progression, input method, child-friendly design, feedback, accessibility and ethics.	Three experts, one session; expert opinion, not child observation; the experts advised confirming details with occupational therapists and educators.
SRC-006	TA meeting	Oct 7, 2026	Issue #78	State machine for the formal specification; define jargon; named TBD constants; specialists as de-identified secondary stakeholders.	Course guidance, not user needs.
SRC-007	Peer review	–	Issues #69–#75	Team 13 review of SRC-001 and SRC-002.	Reviewers' opinions; not all suggestions are adopted.

27.1 Key Interview Statements

This subsection lists the interview statements most relevant to the requirements. Each is a short paraphrase, not a quote; the exact words are in Appendix 28 at the given timestamp.

SRC-005.01 (P1, 28:37; P2, 29:18; 30:20–31:02)

Deciding automatically when a child is ready for the next level would need assessment and is too complex for the time frame. An adult who knows the child (a parent or educator) should choose the level. The team agreed to leave progression to the supervising adult.

SRC-005.02 (P2, 29:18)
In a past project, a teacher personalized each child's activities through a separate adult interface that also showed progress.

SRC-005.03 (P2, 31:42)
Which progress measures help an adult should be decided with occupational therapists or educational specialists, using milestones from the literature if they exist.

SRC-005.04 (P3, 24:34; P3, 25:36; P1, 26:46)
Holding a pen affects how children draw and the usability of the app. Finger input is the easiest start; a larger stylus and then a smaller one could be later difficulty levels if fine motor practice is a goal.

SRC-005.05 (P1, 34:13)
The app should be child-friendly: a child who cannot read yet should be able to enter a lesson, practise, restart and move back and forth using images and icons rather than text, without constant adult help. Adults handle setup.

SRC-005.06 (P1, 36:26)
Positive feedback wording and colours should be designed with feedback from educators and occupational therapists, built up bit by bit.

SRC-005.07 (P2, 37:39)
Audio feedback is a good idea; offering more than one modality is encouraged in inclusive design, with the user able to turn the audio off. Design it carefully with therapists' feedback, and consider adults with ADHD or autism as reviewers when children cannot take part.

SRC-005.08 (P2, 39:26)
Start with the Web Content Accessibility Guidelines (WCAG), then apply more specialized design heuristics for autism from the literature.

SRC-005.09 (P2, 19:05; P1, 20:27; P2, 20:47)
From a universal design view, the skills apply to all children, including those with motor or sight challenges; designing for disability serves many groups.

SRC-005.10 (P1, 13:42; P1, 15:12; P1, 16:13; P2, 16:45)
Camera finger tracing in the air practises gross rather than fine motor

skills. It is useful but should come after the core touchscreen tracing when time is limited.

SRC-005.11 (P2, 22:56; P1, 24:03; P1, 33:00)

Testing with children needs ethics approval (about three to four months where P2 works). Meeting experts for feedback without involving children does not.

SRC-005.12 (41:51–42:48)

All three participants agreed to the use of the transcript for eliciting requirements.

28 Appendix — Specialist Interview Transcript

This appendix contains the transcript of a 43-minute remote group interview held on October 1, 2026, with three domain specialists. The specialists develop software for children with autism, ADHD and other learning needs, and hold doctorates in human-computer interaction and related engineering fields. The interview was conducted by Rayan Nasrallah and Moustafa Moustafa: a 10-minute presentation of the proposed application, followed by a semi-structured discussion. All three participants consented to the use of the transcript for eliciting requirements (see 41:51–42:48 below).

The transcript is de-identified at the participants' request. They are referred to as P1, P2 and P3. Their names, institutions, locations and other identifying details are replaced with bracketed placeholders. A personal health detail is removed. The automatic transcription was corrected where it misheard words, and filler words, false starts and short acknowledgements ("mm", "yeah") were removed. Each turn keeps the timestamp from the original recording (minutes:seconds), which is how requirement sources in this document cite it (for example, P1, 28:37).

Rayan (0:03)

First, I want to introduce myself and also Moustafa. I'm a student at McMaster University. I'm in my final year and I'm studying software engineering with Co-op. So it's an engineering program where we also take a year or so to do an internship program at a company here in Canada. So I'm on my final academic year, and we reached out to you guys because we're doing a project for our final year. It's called the Capstone. And we'll just introduce the project very briefly, and then we want to hear from you guys mostly and get your expertise on the topics we'll be discussing. I'll let Moustafa introduce himself as well.

Moustafa (0:54)

Yes, very similar to Rayan. Also, my name is Moustafa, 4th year software engineering as well. We're in the exact same program. Both kind of had the exact same story. We lived in [location redacted] our whole lives, or for Rayan, it's half of his life, the second half. And then we came to Canada for university. We've been here for five years now. Yeah, Rayan can start.

Rayan (1:16)
Yeah, I'll start sharing my screen. Can you guys see it?

P1 (1:26)
Yes. So both of you are also like final year students in university. Okay.

Moustafa (1:32)
Yeah, we're both on our 4th academic year, but 5th year with the 16 months we took in the middle for the co-op.

P1 (1:39)
Okay, I see.

Rayan (1:41)
Yeah. So last year, we both went for our internship for 16 months, like between our third year and our 4th academic year. So right now we're on our fifth year of university, but final, final year hopefully. Inshallah.

Rayan (2:04)
So I'll just get started. Today, we're going to introduce Squigglr, the application that we're building for our final year engineering project. And over the next 10 minutes or so, we're going to walk you through what we're planning and what we envision for this project moving forward. Let's start with who this app is for. So Squigglr is meant for a pretty broad group of children. That includes kindergarten and primary school kids who are still building their foundational drawing skills and children with autism, ADHD, or other support needs. And one thing we want to be careful about is that we don't want to assume two children with the same diagnosis need the same kind of support. Every child has their own abilities and their own preferences. Furthermore, around each child, you've got the adults who support them, like parents, guardians, and caregivers, educators and learning support staff, and school administrators. We want Squigglr to help them too. Before we jump into the concept of our application, I want to go through some of the research that shaped our thinking. First, when we say visual motor integration, what we mean is coordinating what you see with how you move. For example, seeing a square and then guiding your hand to draw it. The first paper is Daly, Kelley, and Krauss. They looked at kindergarteners and found that kids with stronger visual motor integration tended to be much better at copying letters legibly. So basically, these skills are closely tied to things like

handwriting. It's a correlation though, so it doesn't prove that practicing one improves the other. The second is Taverna and colleagues. They ran a 10-week program with kindergarten and 1st grade children built around short games like completing paths and tracks. And every activity had easy, medium, and hard levels. After the program, the kids' visual motor integration scores went up. There wasn't a control group, so we can't say for sure that the program caused it, but it's a good example of structured practice with graded difficulty, which is pretty much what we're trying to go for. So putting these together, they support the idea behind structured, graded drawing practice, but they don't show that Squigglr itself will get those results. That still has to be evaluated. The next two papers are about why we're paying attention to motor-related needs in children we're designing for. The 3rd paper compared handwriting between autistic children and children without autism. The autistic group had more trouble specifically with forming letters. So basically getting the shapes of the letters right. And within that group, their motor skills predicted how well they wrote. Then the final paper did a systematic review on children with ADHD. They found that fine and gross motor difficulties come up a lot, but the severity ranges from mild to severe. So for us, the big takeaway is really the variation. Not every autistic child or child with ADHD has motor difficulties. And the ones who do are all different. So instead of building one path per diagnosis, we want guidance and difficulty that can be adjusted. And we want kids to be able to practice as much as they need at their own pace. All right, so let's take the first look at the product. Squigglr has three parts connected to each other. First, there are guided activities. That's where kids practice lines, basic shapes, and simple drawings, and they gradually move from tracing to drawing without any guides. Then there's the creative space, where they can just draw freely and use those skills they've built outside of the structured exercises. And lastly, there's practice history, which keeps a record of attempts so parents and educators can actually see what's being practiced. And across all three, you've got adjustable support, history that's saved between sessions, and everything runs in the browser. Now, Moustafa will go through each of these, starting with how an activity progresses.

Moustafa (6:03)

Yes, thank you, Rayan. So we're going to be using the activity of learning

how to draw a square as our example here. First, the child would see the demonstration, so where to start and which way to actually go to draw it. Then they trace over a visible outline. After that, we can start taking guidance away step by step. So maybe they only get the corners or part of the outline. And finally, they would draw the square completely on their own with no guides at all. Kids can repeat any of these stages as many times as they want, and the support can be turned up or down. Now, one thing we haven't done yet is decide exactly when a child moves from one stage to the next. So we're not setting any accuracy thresholds or a required number of attempts. We think that's something that should be informed by specialists, not just [inaudible]. So on the next slide, this is a concept of a tracing activity in progress. So in this example, you can see the child has traced three sides of the square, but they missed the right side, for example. Instead of failing the attempt or just finishing it for them, the app highlights the part that's unfinished and gives a short prompt so the child can go back and complete it in the same attempt. There are two main principles here. First, we keep the child's actual drawing. We never swap in a perfect square and count that as them doing it on their own. And the support can be adjusted too. For example, how much of the outline is shown, or if it's just the corners, as mentioned on the slide before. Second, the feedback is based on predefined observations that we are manually deciding. We're not proposing an AI model that gives personalized advice on each drawing. It would just be basic math and algorithms to figure out what's wrong with the shape they drew and comment on it if needed. However, the exact implementation details are still a work in progress. Now, so alongside the lessons, kids also get a free creative space where they can draw whatever they want. In this example on the left, a child has drawn something that looks like a bird. The idea is that the app would take that and tidy it up just a little bit. So basically the lines would get a bit straighter, the curves get a bit rounder, but it's still the child's drawing. We're not swapping it out for a perfect clip art bird. And if the model can tell what it is, like here it's clearly a bird, it can use that to guide the cleanup. But even if it can't, it will still just clean up the lines and curves a little bit. The way we're planning to do it is we send the drawing coordinates to a lightweight AI model, and then it sends it back slightly adjusted, like it sends back the same thing with slightly adjusted coordinates at the same scale, and we rebuild the drawing inside the application itself. That way, everything stays editable

so the child can still move things, resize them, and change the colour or erase them, and they can always restore their original drawing, so they stay in control of their own work. This only happens in the creative space, though. In the structured activities that we talked about in the previous slides, we always keep the child's real drawing. The main goal for this creative drawing area is for it to be fun and to give kids a chance to use their skills that they learned outside of the exercises. Now here is the concept of what a parent or educator might see on our app. For every structured attempt or exercise, the record includes the activity and its guidance level, the child's drawing, any difficulties they faced, and the final outcome. So for example, in attempt one, the child missed 2 sections at first. But when they went back and completed them, the record keeps both of those. Attempt 2 was the same outline guidance, nothing was missed, and then a later attempt, so attempt 3 near the bottom, it said that was completed with fewer guides and it was fine. Adults can look at which activities were practiced, attempts over time, what guidance was used, and the difficulties that kept coming up or got resolved, and the actual drawing of the child. And there are some things we're deliberately leaving out. So no scores, no comparisons with the general population, no diagnostic labels, and no recommendations for special classes or treatment. We also don't want to suggest that fewer attempts automatically means better visual motor integration. The whole point is to just make practice easier to understand and to support better conversations about what a child needs. So now moving on to how we're thinking of building it, Squigglr is planned to be a web application. Basically, a child practices and draws on a compatible device that supports drawing, like an iPad or a tablet. Their attempts get saved to an account, and then adults can review that history later from any device. Part of the reason we went with the browser is that installing separate apps on school-managed devices can be pretty difficult, according to our supervisor who has some experience working with schools in Canada. We'll be targeting supported browsers and compatible devices, but we're not promising offline access or that it will work on every device just yet. Okay, so just to sum up the core idea, we have structured practice with lines, shapes, and simple drawings, progression from tracing to drawing without guides, guidance and difficulty suited to each child, free drawing with automatic refinement, accurate records of attempts and difficulties, and a clear practice history for parents and educators. There are also two extra features that

are not part of the main requirements, but we wanted to get your feedback on them as well. The first is camera-based finger tracing, where the shapes and guides show up over a live camera view, and the child traces them with their finger in the air. So instead of having to draw on the screen, they would just do it like this in the air, and it would work the exact same way. The second is taking the same approach of the guided practice we talked about for the shapes and the drawing, but applying it to letters as well. So our main vision is a welcoming place for children to practice drawing at their own pace, with a clear picture of that practice for adults who support them. So that's the whole app idea in summary. Thank you so much for your time. And before we get into our questions for you, please let us know if you have any questions or if anything we chose so far wasn't clear.

P2 (12:07)
Thank you very much.

P1 (12:07)
So thank you very much. Actually, I have a couple of questions. It's a very nice application. One of my questions, actually I have three questions. The first one would be, did you guys co-design this app with educators or therapists or people, like basically the end users who would ultimately utilize this application?

Rayan (12:27)
Moustafa, should I answer?

Moustafa (12:29)
Yeah, go, yeah, for sure.

Rayan (12:31)
So basically, like, we just started this project. We haven't built anything. We haven't even come up with the requirements. So our first kind of thing that we wanted to do is reach out to specialists like you guys that have expertise building apps like this. So the whole point of the engineering project is like we actually build a detailed requirements document over the next month or so. And then we start with the implementation towards December and hopefully finish it towards February and then do a usability test before the end of the second semester, before the end of April. So

we're still building the application, like we're still getting the requirements and everything. And the problem with working with kids, I'm sure you guys know, is there's like ethics involved in everything. So our strategy for eliciting requirements was more to go and ask people like yourselves and also like parents and caregivers and so on. So we haven't worked with kids or anything or educators yet, but like this is our first kind of step, if that makes sense.

P1 (12:42)

Okay.

Moustafa (13:41)

Yeah, and also just to add a small note, we also have our supervising professor for the Capstone Project, who also has some experience in this field. So he's partnered with a couple of organizations in Canada, and he has extensive experience with working with kids with autism and stuff like that. So he's going to be able to help us out as well.

P1 (13:42)

Okay. Okay, sounds great. I don't know why I thought that maybe this app was already being developed. The second question I have is a question that you actually proposed in the last slide about the camera, whether they should trace it using the camera. I mean, the camera-based finger tracing, exactly. This idea is nice, but I would say from my viewpoint, I think it might target a very different task. So whenever they're doing it in the air, that might be more of like a gross motor skill. And then whenever they're drawing on an iPad, that's more on the fine motor skills. So this is something that you could probably check with, you know, either therapists or educators and things. But this is my feedback on that. And you could integrate both. I mean, the camera-based finger tracing is more kind of like an augmented reality component that you would be including. And it's very interesting, especially with children who are more neurodiverse. But you just want to focus; they might be targeting different aspects.

Moustafa (14:55)

Okay. Okay, makes sense. But do you think it might be a good idea in general though to have it? Like, do you think there's actually something helpful about this feature if we do end up implementing it?

P1 (15:12)

I do, because a lot of times, even just looking at child development, before they start the very fine tracing, they usually have big markers, big things, and they're kind of tracing it very largely. So this would replicate that. And that kind of works more on the visual motor, even if it's not fine motor, it's more on the gross motor skills. And it helps them also in the visualization. So if they can visualize tracing a square, it might make it easier for them to, you know, trace it when it comes on an iPad and utilize their fine motor skills. So I think it's definitely an important feature to incorporate.

Rayan (15:48)

For now, though, because we might have a limited scope, like we have a certain amount of time to implement this app, do you think our focus with the shapes and the tracing and teaching them to draw is a good starting point and that we should focus on that before implementing this camera feature? Or do you think it should be part of this application from the get-go?

P1 (16:13)

I mean, if you're limited in time, I would say just focus on the first part, and then you could progress later on to extend it to the camera-based finger tracing.

Rayan (16:23)

Sounds good. Thank you for the feedback.

P1 (16:28)

I have a couple more, like some feedback, but I'll give maybe [P2] an opportunity if she has any questions and then I can continue.

P2 (16:40)

Thank you. It's really very nice.

P1 (16:43)

Your voice is very far, [P2].

P2 (16:45)

Can you hear me now? Can you hear me well? So it's really very nice work, both. It's just very inspiring what you're doing. So I agree with [P1]

in terms of focusing and leaving the camera and the tracking for maybe something to extend beyond this project. That's one thing. A second aspect, actually, while you were introducing it, I was thinking about my own self when I was very young in elementary school. I've always had difficulty in drawing a line. So when I saw this project, it's very nice. I think it's a very relevant problem. That's first. And my question to you: are you targeting specific groups of children with learning needs, or is it kind of broader? I would like to hear from you, who is your sample population?

Rayan (17:45)

So the idea started for kids with special needs. So kids with autism and ADHD, because the whole thing is each case differs. But from what I've seen personally, even as kids tend to get older, they still struggle with these basic skills that really young kids like toddlers and kids in kindergarten learn to develop early on in their lives, right? So our idea is we're mainly focusing on kids with autism and ADHD, but we also want to extend it to very young children, right? Like this is a skill that, like you said, you remembered when you were young. A lot of kids don't practice this when they're young. It's not integrated into learning. And I feel like it is a very necessary skill. So not just kids with special needs that are a little bit older, but also very young kids as well. We think they can use this application. And one big component of our design, which we wanted to ask about later, was how we can let the guidance and the support be controlled by the caregiver or the parent that's there supporting the activity. Or even if we should have a parent or a caregiver supporting the activity in the first place. So that's where it goes on to.

P2 (19:05)

Thinking about it from an inclusive, universal design perspective, this could be relevant to, yes, I agree, autism and ADHD, but then also children with gross motor and fine motor challenges, and also kids with sight challenges. [Personal account of a childhood vision condition removed.] So I believe this could be a training tool for any child in need of it, regardless of their situation, because there are many conditions that could affect our visual perception, and also affect even our handwriting. So just like when I saw this, I felt like...

Rayan (20:13)
Yeah. That's the plan, hopefully.

Moustafa (20:14)
So similar to what Rayan said, you don't think we should only limit it to children with special needs or autism or ADHD or anything like that? It should be general.

P2 (20:14)
Yeah, because...

P1 (20:27)
Yeah, I agree. I don't think you should limit it, because the skills that you're trying to teach now are applicable to all children, right? But I guess depending on their disability, they might achieve it at different stages in life. But if you go to nurseries that are looking at children at two and three years old, this is what they're focused on.

P2 (20:47)
Exactly. That's true. Actually, there is a saying in the world of accessibility: if you design for people with disability, you are actually designing for all. So whatever serves a certain population could be of use for many other populations. Then, thinking even of elderly people, people who get a stroke at some point in life, this is also very effective. I would see it in a rehabilitation centre, for example.

Moustafa (21:18)
Yeah, I didn't think of it, yeah, that's true.

P2 (21:25)
So the use cases could be endless, really. And I think that's also, if it's a startup, a selling point as well, because it could be put in a rehabilitation centre, it could be put in a kindergarten, it could be put in a school or at home.

Rayan (21:45)
Inshallah.

Moustafa (21:47)
Something happened to your mic. Yeah, yeah. I think it disconnected.

P2 (21:57)
We can see you, but we can't hear you.

Rayan (22:00)
Hello!

Moustafa (22:03)
Talk again.

Rayan (22:04)
Okay. Can you guys hear me?

Moustafa (22:06)
Yeah, that was it.

Rayan (22:07)
Yeah, yeah, I forgot what I was saying. Yeah, we wanted to ask actually if at some point we are able to build this app, would you guys, because you guys actually do research, would you guys see yourselves kind of looking into this area, or using this app, or doing further research? Like, if we just start with the idea, like a proof of concept, this app could serve as a proof of concept. Would you guys be able to take it beyond that in the future? And hopefully maybe in February, be able to actually get children to use it and do some observation, not necessarily a trial or research, but just observation activities, because it's difficult for us to do it here in Canada. That's why I wanted to ask.

P2 (22:56)
Yeah, but actually, it's equally difficult [here]. You need to get the ethical approval and the process is quite... Yeah, I think the laws have been tightening over the last 10 years, I would say, especially in this part of the world and in research. But it's easy for us to obtain approvals because we have experience, and an approval would take about three to four months really to obtain. But then actually, as you also speak, I am aware of a group at [another local university] that is looking at handwriting and dyslexia and other learning [needs]. So there is also a very active group there that looks at learning needs and technology, the use of image processing in particular. I can share their contact with you if you're also interested to meet more expertise in the field. And I also

wanted to say something else. Well, I forgot, but these are the two points from my side, but this is really nice, very nice work.

Moustafa (22:59)
Yeah, I think it's difficult everywhere.

Rayan (24:03)
Thank you.

Moustafa (24:03)
Thank you so much.

P1 (24:03)
Yeah, absolutely. I think it's great work. I think it's much needed. And I mean, we would be happy to kind of help guide you. I think there is some merit in trying to, you know, pilot or test this application. Of course, you would need an IRB, but maybe you could do this where you're at, through your university, right? And we could support you through the process and kind of help guide you. So I think it's a great application.

Rayan (24:26)
Yeah, that would be perfect. Yeah. Thank you guys. Appreciate it.

P2 (24:31)
Thank you.

Rayan (24:32)
[P3], you.

P3 (24:34)
Yes, I just want to ask one thing. Honestly, I'm curious, I'm not sure if I missed it or not while you were presenting. Will you ask the children to hold the pen in order to do the line, like the drawing, or will it be their finger? Like, will it be touch, or are you thinking of a pen?

Rayan (24:56)
I think it can be either. We haven't really got into that decision. Our initial thought was a pen. Like, I have this iPad here. So like this, for example. And then maybe as well, there's attachments you can buy for these that make the grip wider. Because I know some kids have trouble holding it like this. So I know there's grips that you can buy that make

it large. So it's very flexible. I feel like it should support tracing with the finger and pen. But what do you think?

Moustafa (25:03)
Thinking about it now, a pen is probably not a good idea, right?

P3 (25:11)
Exactly, yeah.

P3 (25:36)
And that's why I ask, because as [P1] was saying and [P2] also was saying, regarding the fine motor skills of the children, like how they will hold the pen? Will they hold it from the top, or some will hold it like this? You know, I don't have a pen near me, but they will hold it like this. So this might affect how they will draw, which might affect maybe the usability of the app itself. So if your goal is not about the fine motor or something, we could remove the pen. If your goal is more about the drawing itself, only the drawing idea, I guess you can use the finger thing. Not sure if [P2] and [P1] agree with me on this idea, but just to make your app more reliable in real life for the children.

P1 (26:35)
So you mean to draw with their finger on the application?

P3 (26:40)
Yeah, it will be like a touch. They will draw as Moustafa was saying in space using the camera. So it will be the same but on the tablet or iPad itself, rather than holding a pen and having more challenges.

P1 (26:46)
In the air, yeah. That may be a start. I see that. That's a good option for a start. But again, if the idea is to kind of improve the fine motor skills, they have to practice with being able to hold something. So usually they start with something large, because it's easier for their hands, and then they have to go to the smaller ones, like pencils. That's more difficult, so that actually takes some training for them to do. So you could embed difficulty levels, you know, where they start with their finger, which is the easiest, and then with a larger kind of marker or pencil, and then something smaller that's more reflective of what they would use in schools.

Rayan (27:31)

So on that point, we had a question about how we should adjust, because in our thought process, we don't want the app to judge when a child is ready to move from one stage to the next. So the example we gave throughout the slides was the square. But of course, a square is a very simple shape. And then they could move to triangles and then stars and more complex objects that they can trace and then draw by themselves, right? And we'd have an accuracy threshold. But our concern was, should we dictate when the child is ready to move from a square to a triangle or a square to a star? Or is that something we should leave fully to the supervising adult during the activity? And what determines when a child is ready to move on? Because now you introduce an extra degree of complexity, which is holding an object in and of itself; decreasing the size of that is also progression, but that's not something we can necessarily track.

P1 (28:37)

So if you're going to look at that, then you're going to open up a door of assessments, right? Because there needs to be assessment. And I think that, given your time frame, it might be too complicated at this stage. I would keep it user interactive, so the user can decide whether they're going to go from, I don't know, level one, level 2, level 3, right? And then ultimately who decides what they're going to do would be either their parent or the educator or somebody who knows where their level is at. And then they could tell them, you know, you need to practice level one until you're proficient in it, and then they can ask them to move. But if you want the app to do it automatically, that's going to be more advanced.

P2 (29:18)

Actually, just something we had in one of our projects: we had an app for children to learn vocabulary, and we co-designed it with a centre [location redacted] with speech therapists in the centre, and we allowed the teacher to kind of personalize the journey for each child separately, and that's not really automated. The teacher would have a back-end kind of interface to do that. But as [P1] said, that actually will bring you to the other side of the story, which is assessment. But then you might want to think about just allowing the teacher or the caregiver to kind of change the activity

and giving them a back-end view of the progress. But then, thinking also, if you do that, then one would ask, oh, what kind of assessment are you applying here?

Rayan (30:20)

Yeah. So I think maybe the best thing is to do what you guys recommend, which is to kind of leave it to the caregiver or the supervising adult, and we shouldn't worry about assessment right now. It shouldn't be in the scope given the time frame. Maybe something in the future, Moustafa, I don't know if you agree.

Moustafa (30:38)

Yeah, I feel like it's very good to have, but maybe not in these short months. It would be nice, but I don't think we have time to do it right now. So right now, as you guys recommended, it would probably be the adults who would say, okay, this is fine for now, move on to level 2 or level 3, instead of us having to figure out how to map when this child is ready for level 2 or level 3. We can figure that out later.

Rayan (30:52)

Hundred percent, yeah.

Rayan (31:10)

Speaking of that, in the history that we showed, what statistics or what values or what information would actually be useful to the parent or caregiver, so they can say, okay, this progress history is actually benefiting my judgment of whether the child is ready to progress? For example, should we have a percentage, like, okay, the accuracy threshold is above 80%, we passed it, but we're still showing you that this is the accuracy of the kid's drawing?

P2 (31:42)

That's a very good question. Really, I think this is a question to ask the therapists or the teachers, if there are certain defined milestones in the literature that you could kind of have as your benchmark or reference there. And that's something to discuss really with, I think, either an OT, an occupational therapist, or an educational specialist.

Rayan (32:12)

Yeah. Do you guys know any occupational therapists or educational

specialists that we can speak to about something like that, that you could connect us with?

P2 (32:21)

I think the easier way to do that is reaching out to private centres [location redacted]. I think [P3] can share them with you step by step. They are very good, they are very approachable. You might want to introduce this, and they are very interested in technologies. So, maybe that.

P1 (32:42)

And [a local hospital] as well. They have a therapy program where they have occupational therapists as well.

Rayan (32:50)

We're scared that if we email them, we might scare them or something. They might start talking about ethics and stuff, but it's just a small project. It doesn't seem like it, I guess, when we reach out to them, but it's hard to kind of...

P1 (33:00)

Well, the ethics only comes in whenever you say that you want to test your product, you want to meet with the kids and stuff. But if you're asking just to meet with them and get their expertise and get their feedback, it should be okay as long as you're not really interacting with the children at this point. Just to get feedback through a meeting, like what we're doing now, it should be fine.

Rayan (33:26)

Okay. Sounds good. Moustafa, do you have any more questions you wanted to ask?

Moustafa (33:34)

Yeah, this was on a point that we were mentioning before. So with deciding when the child goes on to the next level and stuff like that, a similar question: how should this app be made? Should this app be built in a way that the child by himself can go in and kind of know how to play around and move from level one to level 2, or even go to the creative mode and draw? Or are we expecting that there's always a parent present? Because we've never worked with kids with disabilities, or not even kids with disabilities, just children in general, very young

people. We don't know how to give feedback, like, oh, good job, now try this. Or how do we tell them to draw a square? Or how do we tell them to do this? Or are we assuming that there's a, yeah?

P1 (34:13)

That's a very good question. Yeah, so to answer the first part of your question, I think you should design the application so that it's child-friendly. Because if you're going to design it in a way that the parent has to keep, you know, interfering, parents are going to quickly get annoyed, and, you know, if they give them the iPad to practice, they want them to practice without their constant interference. So then you have to think about how you design the application so that it's child-friendly, meaning that, you know, both the child and the adult can understand the interaction. So a child who's three years old maybe doesn't know how to read yet, but, you know, kids these days are very tech-savvy, so they know how to navigate applications and things. But if you look at the games that they play, it's usually not very text heavy or oriented, so they have images and icons that they navigate in order to go and play. So you would have to design it from the get-go in that kind of manner. Adults obviously will know how to navigate through and install and do the initial parts, but you know, you should also design it in a way so that children can navigate and be able to enter the lesson that they're on, be able to practice, to restart, to go back and forth. Because otherwise, if you make it too high level, where they require constant adult supervision, I think you're going to limit the usability of it.

Moustafa (35:31)

Okay, so it should be somewhat expected that the child can use it alone, play around with it.

P1 (35:36)

Yes, and then, yes.

P2 (35:36)

I think also, sorry, [P1]. It's just, I think also it's your decision on what you want the app to look like. Like, you know, there are many drawing apps in the App Store, right? For children, for mainstream, or maybe not even in mainstream school, it could be used in a centre. But it's your decision to say whether you want this app to kind of follow a process and

have, yes, a child as a user, but then the caregiver can give them the access. So I think these different workflows, you can put whatever you like, because there is no right and wrong, sort of. There's only the way you present it and the angle in which you're presenting it.

Moustafa (36:22)
I see, I see.

P1 (36:26)
No, no, there's no problem at all. I think the next step would be for you guys to meet the end users and the stakeholders, which would be the educators and occupational therapists. They can also give you more feedback. And they're the ones who will be able to guide you, because we're developing similar, not similar, but I mean, similar types of applications, one for occupational therapy. And even the way that we give them positive reinforcement, like if they get something correct, what we say and how we say it and the colours that we use. I mean, we go through the literature, of course, but we also get the feedback from, you know, the experts, because we're doing the design and the technology aspect, but they're the ones who deal with the patient side or the children's side on a day-to-day basis. So these kinds of design aspects, you can build bit by bit.

Moustafa (37:15)
I see. Okay, thank you so much. Rayan, you had something to add.

Rayan (37:18)
I just had a follow-up, yeah, so... The first thing is, what do you guys think about voice, like dictation, like voice support for the kids? Like saying, oh, you did a good job, with like some sound? Or do you guys generally see that that's not a good idea?

P2 (37:39)
No, actually, it is a very good idea. We had an app, the vocabulary learning one, that had a voice input as well. And multi-modality in general, like having two modalities to communicate, and then the user could choose; he could turn off the audio. So multi-modality is very much encouraged in the field of design, right? If you want to design an app... now if you look at the Apple products and the phones in particular, they

also have the haptic modality, so the phone can shake sometimes if there is a notification. So multi-modality is always encouraged, even in the realm of inclusive design and universal design. It's always encouraged, but it's just, as [P1] said, designing it carefully, getting the feedback from the therapists, and even, if you have the chance, getting the feedback from the users themselves. Even if you cannot reach the children as a sample or a group, you could reach an adult group with ADHD and autism. And if you want to expand it to different adult groups, then that's also useful, because then you based your decision on real experiences.

Moustafa (38:58)

It's a very good idea, okay.

Rayan (38:58)

Okay. And then I have a follow-up to that as well. In general, what design choices in apps tend to cause problems for kids with autism and ADHD? And also, what are best practices for these types of apps? Because you all have worked on HCI and these types of apps, making sure the interface is specific to their needs. So what general patterns do you guys recommend we should follow?

P2 (39:26)

So yeah, this is a very good question. I teach an elective course for the undergrads in the spring on ICT accessibility, so how to design accessible interfaces and apps, and the main set of guidelines is called the Web Content Accessibility Guidelines. We call it the WCAG. And it's very famous. It actually is adopted as law in many countries, including the US, and [location redacted] adopted it last year. So I think start with the WCAG and then move forward to more specialized guidelines too. Start with the WCAG and then look at design guidelines for autism in particular; we call them heuristics. So heuristics for autism, app design for autism. There is a lot in the literature; just write in Google Scholar "heuristics for app design for autism" and you'll find a lot. But I would suggest starting with the WCAG and then moving to the more detailed ones.

Rayan (40:39)

Okay, perfect. Thank you. Thank you.

Moustafa (40:39)
Your course would be very helpful for us right now, the accessibility course. I wish, I wish.

P2 (40:44)
Yeah, you're welcome to attend, welcome to come [location redacted].

Rayan (40:44)
Yeah. Yeah. Hopefully, that's the dream. Inshallah. Do you guys have any more comments about that? Like, [P1], [P3]? Do you guys have any advice or any patterns that you guys think of?

P1 (41:08)
I mean, from my end, no. I think what you're doing is really great and it's very timely, and we would be happy, maybe after a couple of weeks or a month, to meet again. And if you have questions, we can again give you our view and feedback, just to kind of help guide you through the process.

Rayan (41:26)
Yeah, that would be perfect. I mean, we have a lot more questions, but because we're running short on time, would it be okay if we sent you guys a form or something that you can answer on your own time, whenever you have time?

Moustafa (41:26)
That would be amazing, yeah. It would probably be just very, very short questions.

Rayan (41:44)
Like, they're very specific.

P2 (41:49)
Yes, of course.

P1 (41:51)
Yeah, that's fine.

Rayan (41:51)
Okay, perfect. Well, we'll create a form then, and we'll send that out to you guys, and it has some more questions that we didn't really get to ask.

And also, the whole purpose of this interview, like we said, was to kind of get your feedback and then elicit requirements from that. I guess now I'm realizing we have a lot more work to do. We have to talk to some more people before we can actually elicit the requirements properly. But I was going to ask you guys if you could sign a form, a consent form, that we can use this transcript to basically do that work. We'll send it to you guys as well, with the questions, at the same time, by email, if that's fine.

P2 (42:37)
Yeah, of course.

P1 (42:38)
No, it's fine.

Moustafa (42:39)
Yeah, it's basically just saying that we can use the transcript of what we talked about here to elicit the requirements, and that we had this meeting, basically. That's all.

P3 (42:40)
Sure, no problem.

P1 (42:47)
Sure.

Rayan (42:48)
Okay, perfect. Thank you guys so much.

Moustafa (42:49)
Thank you so much. Also, does anyone have any other questions before we end?

P1 (42:52)
No problem.

P3 (42:56)
Thank you so much. That's amazing work.

P1 (42:58)
No problem. Best of luck [inaudible].

Rayan (43:00)
Well, we'll be reaching out to you guys again, hopefully in the near future. Inshallah.

Moustafa (43:00)
Thank you guys so much. This was very helpful. Inshallah.

P3 (43:06)
Inshallah, Inshallah.

Moustafa (43:08)
Thank you so much. Thank you so much.

P1 (43:09)
Thank you.

Rayan (43:10)
Thank you guys for your time. Bye-bye.

P3 (43:11)
Thank you, bye.

P1 (43:14)
Bye.

29 Appendix — Generative AI Use Disclosure

[ChatGPT (version 5) was used to brainstorm ideas for the template for this disclose section. —SS]

29.1 AI Tools Used

[Give the name of the AI tool(s) used and the version. —SS]

29.2 How and Where Used

[Explain what the tool(s) were used for. Examples include: brainstorming, explaining a technical concept, reviewing the document, generating or modifying text, generating or modifying diagrams, identifying errors or omissions or inconsistencies. —SS]

[For each of the uses, identify which parts of the document were affected. —SS]

[You do not need to report every interaction, but you should disclose the uses that materially contributed to your work —SS]

29.3 Verification of AI-Generated Content

[

- Describe how the team checked AI-generated or AI-assisted material before including it in the document.
- In particular, verify factual claims, technical assertions, requirements, calculations, references, and other information for which an AI system may produce incorrect or fabricated results.
- **The team is responsible for the accuracy, completeness, consistency, and appropriateness of the submitted document, regardless of whether material was generated by a human or an AI system.**

—SS]

30 Appendix — Reflection

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?

2. What pain points did you experience during this deliverable, and how did you resolve them?

3. How many of your requirements were inspired by speaking to your client(s) or their proxies (e.g. your peers, stakeholders, potential users)?

4. Which of the courses you have taken, or are currently taking, will help your team to be successful with your capstone project.

5. What knowledge and skills will the team collectively need to acquire to successfully complete this capstone project? Examples of possible knowledge to acquire include domain specific knowledge from the domain of your application, or software engineering knowledge, mechatronics knowledge or computer science knowledge. Skills may be related to technology, or writing, or presentation, or team management, etc. You should look to identify at least one item for each team member.

6. For each of the knowledge areas and skills identified in the previous question, what are at least two approaches to acquiring the knowledge or mastering the skill? Of the identified approaches, which will each team member pursue, and why did they make this choice?